

Distributed Systems MP1 Group23

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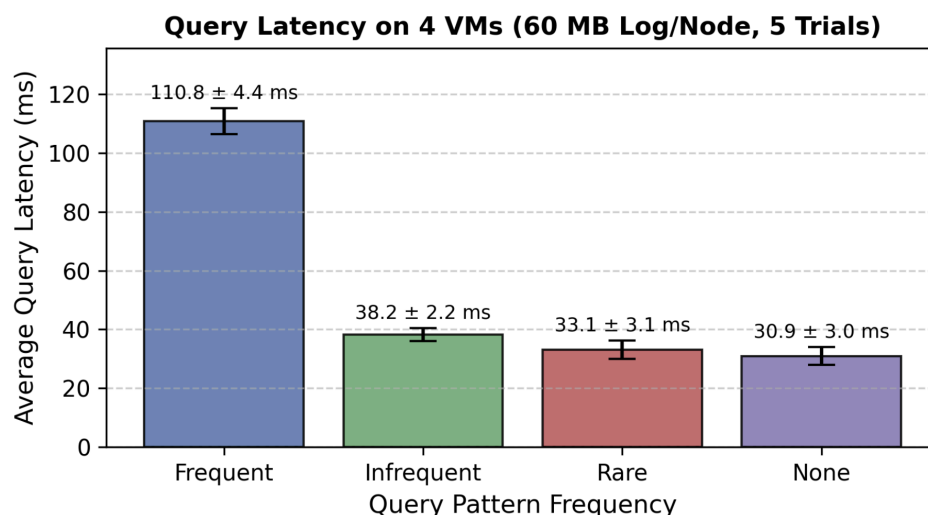
Design & Architecture:

We designed the distributed grep using Go. The client dispatches the query request concurrently to all nodes. Then each node runs a TCP server executing real grep commands against its log file. Finally, the client aggregates matching lines, line counts, and tracks node reachability with timeout handling.

Unit Testing:

We implemented an automated test across 6~10 randomly-chosen VM clusters. It validates exact line and match counts for different frequency patterns and complex regex queries. The test simulates a single-node failure by terminating the server process; it collects complete results from the healthy nodes while marking the failed node as unreachable.

Query Latency & Discussion:



- Frequent (~100,000), Infrequent (~10,000), Rare (~1,000), None(0)

We measured the query latency (end-to-end time until the last node finishes responding) across 4 VMs storing 60 MB logs each over 5 trials.

Analysis:

Latency increases with query frequency. Scanning the fixed 60 MB file in the OS introduces a baseline cost of ~30 ms (observed across zero-match queries). As match frequency rises, data serialization, socket buffer overhead, and network transmission time dominate, leading the frequent query latency to 110.3 ms. Standard deviations remain small (≤ 4.4 ms), confirming network stability across the cluster.